

3.4 Hydroxy Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.4 Hydroxy Compounds
Difficulty	Hard

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

Four reactions on butan-2-ol are shown below

- Reaction 1: Butan-2-ol $\xrightarrow[\text{Heat}]{\text{H}_3\text{PO}_4}$ compound **A**
- Reaction 2: Butan-2-ol $\xrightarrow{\text{reagent X}}$ compound **B** + HCl + SO₂
- Reaction 3: Butan-2-ol $\xrightarrow[\text{room temp}]{\text{PBr}_3}$ compound **C**
- Reaction 4: Butan-2-ol $\xrightarrow[\text{room temp}]{\text{reagent Y}}$ 2-chlorobutane

Compound **A** is a stereoisomer of cis-but-2-ene.

i)

Draw the structure of the stereoisomer of cis-but-2-ene.

[1]

ii)

Explain why compound **A** is less soluble than butan-2-ol.

[2]

[3 marks]

Question 1b

The reaction to form 2-chlorobutane produces three products.

i)

State the name of reagent **Y**.

[1]

ii)

Give the formulae of the **two** other products formed in this reaction.

[2]

[3 marks]

Question 1c

Give the equation for the overall reaction between butan-2-ol and phosphorus tribromide which occurs at room temperature.

[2 marks]

Question 1d

Draw the structure of compound **B** and identify reagent **X**.

[2 marks]

Question 2a

The structure of glycolic acid is shown in Fig. 2.1.

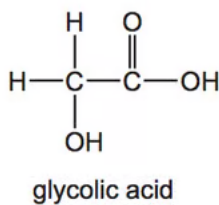


Fig. 2.1

Complete Table 2.1 to show what you would observe when an aqueous solution of glycolic acid is added separately to each of the reagents. If a reaction occurs, state the functional group of glycolic acid that is responsible for the reaction.

Table 2.1

Reagent	Observation with glycolic acid	Functional group
Mg (s)		
2,4-DNPH		
Acidified KMnO ₄		

[3 marks]

Question 2b

Glycolic acid can also be made by reacting glyoxylic acid with NaBH_4 as shown in Fig. 2.2.

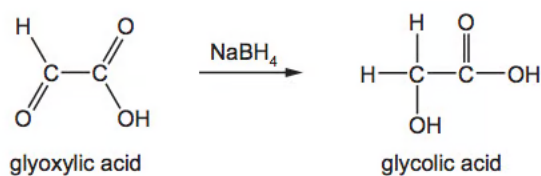


Fig. 2.2

i)

State the role of NaBH_4 in this reaction.

[1]

ii)

Write an equation for this reaction using molecular formulae. Use $[\text{H}]$ to represent NaBH_4 .

[2]

iii)

Explain why NaBH_4 is a suitable reducing agent for this reaction.

[1]

[4 marks]

Question 2c

When glycolic acid is heated in the presence of a sulfuric acid catalyst, a new compound, **Y**, $C_4H_4O_4$, is formed. The equation for the reaction is given.



The infrared spectrum of **Y** is shown in Fig. 2.3.

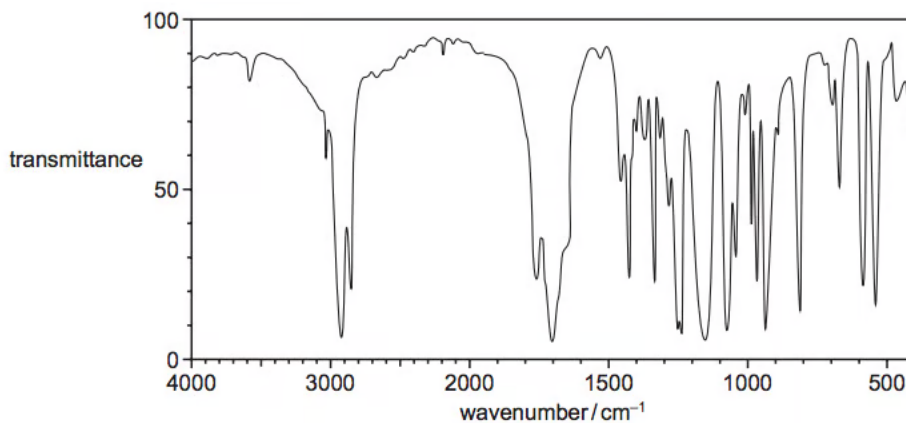


Fig. 2.3

i)

State how this spectrum differs from an infra-red spectrum of glycolic acid. Using Table 2.1, explain your answer with particular reference to the peaks within the range $1500 - 4000 \text{ cm}^{-1}$.

Table 2.1

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumber) / cm^{-1}
C–O	hydroxy, ester	1040 – 1300
C=C	aromatic compound, alkene	1500 – 1680
C=O	amide	1640 – 1690
	carbonyl, carboxyl	1670 – 1740
	ester	1710 – 1750
$C \equiv N$	nitrile	2200 – 2250
C–H	alkane	2850 – 2950
N–H	amine, amide	3300 – 3500
O–H	carboxyl	2500 – 3000
	hydroxy	3200 – 3600

[2]

ii)

Suggest a structure for **Y**.

[2]

[4 marks]

Question 3a

Compounds **W**, **X** and **Y** are all straight chain carbohydrates with **X** and **Y** each containing five carbons. Compound **W** and a by-product, compound **Z**, are formed from the reaction between compound **X** and **Y**.

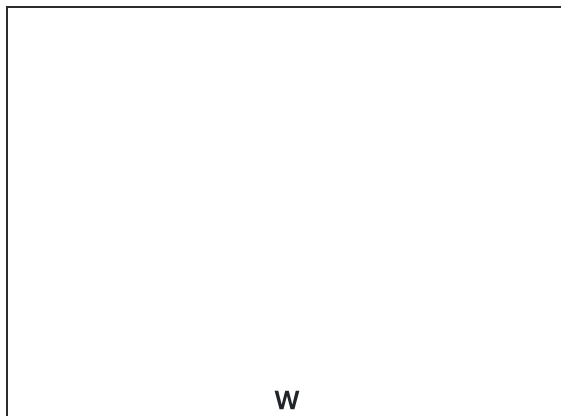
Compound **X** can be fully oxidised by the reaction with acidified potassium manganate(VII) to give compound **Y**. 5.508 g of compound **Y** contains 0.054 moles.

State the name of compound **Y**. Justify your answer.

[3 marks]

Question 3b

i)

Draw the full displayed formula of compound **W**

[1]

ii)

Explain how compound **Z** is formed in the reaction

[1]

[2 marks]

Question 3cCompound **X** will oxidise to compound **Y** if allowed to fully oxidise.Explain how a student could stop the full oxidation of compound **X**.

[4 marks]

Question 3d

i)

Write the structural formula of an isomer of compound **X** that will **not** react with acidified potassium manganate(VII).

ii)

Using your knowledge of bonds present in the oxidation products of different classes of alcohols, suggest why the isomer you have drawn will not react with acidified potassium manganate(VII).

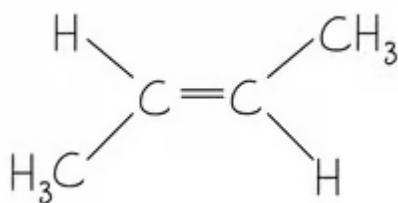
[5 marks]

Model Answers: Hard

1a

a)

i) The structure of compound **A** is:



- Correct structure of trans-but-2-ene; [1 mark]

ii) Compound **A** is less soluble because:

- It does not contain OH / hydroxyl group(s)

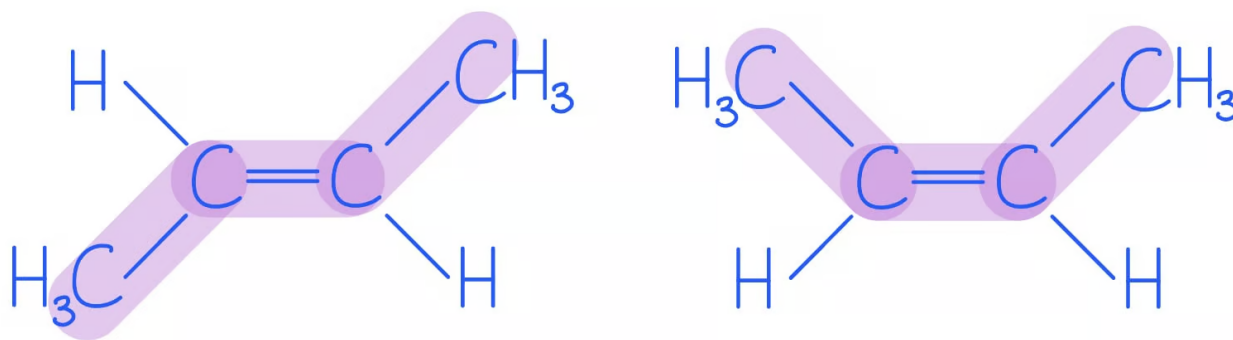
OR

It is not an alcohol; [1 mark]

- So compound **A** / it does not form hydrogen bonds with water; [1 mark]

[Total: 3 marks]

- This question asks you to draw compound **A** which is an isomer of cis-but-2-ene
 - **Careful:** This could easily be configured to draw the structure of cis-but-2-ene so double-check the question!
- The question is asking for the stereoisomer of cis-but-2-ene which is actually **trans-but-2-ene**
- Trans and cis but-2-ene are shown below



1b

b)

i) The name of reagent **Y** is:

- Phosphorus pentachloride; [1 mark]

ii) The formulae of the **two** other compounds are:

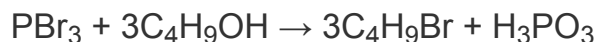
- HCl; [1 mark]
- POCl₃; [1 mark]

[Total: 3 marks]

- **Careful:** The question clearly asks for the name of the compound in part (i) and the chemical formulae of the compounds in part (ii)
 - Make sure that you give the name or chemical formula as asked
 - Do not give the chemical formula in part (i)
 - Do not give chemical names in part (ii)
- The clue in the question is that the reaction occurs at room temperature
 - PCl₃ could also form 2-chlorobutane, but this would require heat for this reaction to occur
 - PCl₅ is a more reactive compound, so will form 2-chlorobutane at room temperature
- The overall equation is:
 - $\text{CH}_3\text{CH}_2\text{CHOHCH}_3 + \text{PCl}_5 \rightarrow \text{CH}_3\text{CH}_2\text{CHClCH}_3 + \text{HCl} + \text{POCl}_3$

1c

c) The overall equation is:



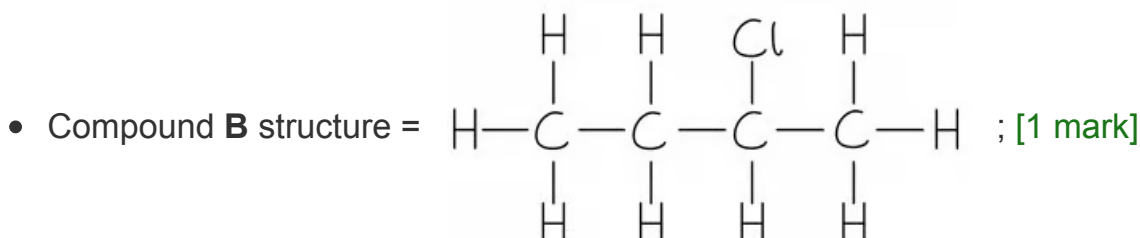
- Correct chemical formulae; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- This reaction is similar to PCl_3 reacting with an alcohol

1d

d) The structure of compound **B** and the identity of reagent **X** are:



- Reagent **X** = SOCl_2 ; [1 mark]

[Total: 2 marks]

- This is another example of a substitution reaction
- The clue to the identity of reagent **X** is the production of SO_2 ,
 - Reagent **X** must be a compound containing sulfur, which should guide you to thionyl chloride / SOCl_2
- The overall reaction is:
 - $\text{CH}_3\text{CH}_2\text{CHOHCH}_3 + \text{SOCl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CHClCH}_3 + \text{SO}_2 + \text{HCl}$

2a

a) The completed table is:

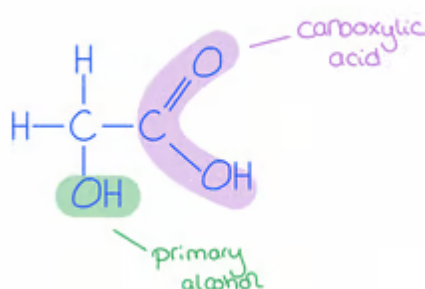
Reagent	Observation with glycolic acid	Functional group
	effervescence / fizzing /	COOH / carboxylic acid; [1

Mg (s)	bubbling	mark]
2,4-DNPH	no visible reaction	(no group required); [1 mark]
Acidified KMnO ₄	purple to colourless	–OH / alcohol; [1 mark]

Observation and functional group **BOTH** required to gain [1 mark]

[Total: 3 marks]

- Make sure you pick out the correct functional groups for each reaction
- It can help to circle the functional groups present



- 2,4-dinitrophenylhydrazine (also known as 2,4-DNPH) is a reagent which detects the presence of carbonyl compounds (compounds with $-\text{C}=\text{O}$ group)
- Carboxylic acids react with the more reactive metals to produce a salt and hydrogen

2b

b)

i) The role of NaBH_4 is:

- Reducing agent / reductant; [1 mark]

ii) An equation for this reaction using molecular formulae. Use $[\text{H}]$ to represent NaBH_4 is:



- Correct molecular formulae for $\text{C}_2\text{H}_2\text{O}_3$ and $\text{C}_2\text{H}_4\text{O}_3$; [1 mark]
- Correct balancing; [1 mark]

ii) NaBH_4 is a suitable reagent because:

- It reduces the $C=O$ / aldehyde group (of glyoxylic acid)

AND

It will not reduce the $COOH$ group (of glyoxylic acid); [1 mark]

[Total: 4 marks]

- $NaBH_4$ is a reducing agent, but there are also other reducing agents that you need to know about
 - They are not a 'one size fits all' set of reagents
- $NaBH_4$ will reduce aldehydes and ketones, but not carboxylic acids
 - $LiAlH_4$ will reduce all of these

2c

c)

i) This spectrum differs from an infra-red spectrum of glycolic acid by:

EITHER

Glycolic acid would have:

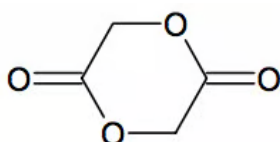
- 2500 – 3000 due to RCO_2H ; [1 mark]
- Range within 3200 – 3650 due to $RO-H$; [1 mark]

OR

Spectrum **Y** would **NOT** have:

- 2500 – 3000 due to RCO_2H ; [1 mark]
- Range within 3200 – 3650 due to $RO-H$; [1 mark]

ii) A possible structure of compound **Y** is:



- Ester group
- AND**

C₄H₄O₄ molecule; [1 mark]

- Correct cyclic structure; [1 mark]

[Total: 4 marks]

- You should quote the data given in Table 2.1
- As glycolic acid has both alcohol and carboxylic acid groups, it can form an ester with itself
 - You would gain a mark for giving any ester group as well as drawing a valid C₄H₄O₄
 - Your structure should be cyclic to gain the second mark

3a

a) To work out the identity of compound Y:

- $M_r = \frac{5.508}{0.054} = 102.0$; [1 mark]
- Compound Y name = Pentanoic acid; [1 mark]

Justification:

- (Compound Y) has the formula C₅H₁₀O₂
AND
Is an oxidation product of compound X; [1 mark]

[Total: 3 marks]

- There is a lot of information given in the question and the first step will be to calculate the M_r of compound Y
- You know that it contains 5 C atoms so that is most of the molecule accounted for
- You are aware that it is an oxidation product so must contain at least 1 oxygen atom
 - In this case, it contains 2 oxygen atoms
- The M_r of pentanoic acid is 102.0 whereas the M_r of pentanal is 86.0
 - Therefore, compound Y has to be pentanoic acid

- Ester group

AND

$C_4H_4O_4$ molecule; [1 mark]

- Correct cyclic structure; [1 mark]

[Total: 4 marks]

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- As glycolic acid has both alcohol and carboxylic acid groups, it can form an ester with itself
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3a

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- $M_r = \frac{5.508}{0.054} = 102.0$; [1 mark]
- Compound Y name = Pentanoic acid; [1 mark]

Justification:

- (Compound Y) has the formula $C_5H_{10}O_2$
AND
Is an oxidation product of compound X; [1 mark]

[Total: 3 marks]

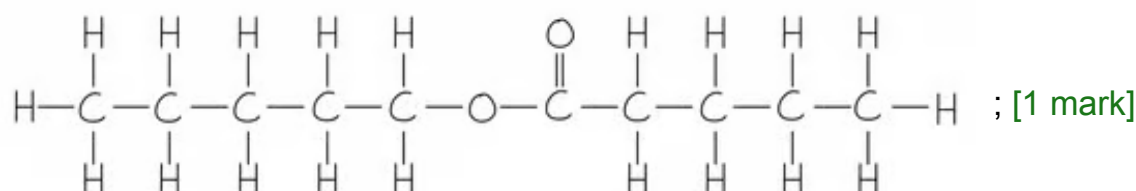
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- The M_r of pentanoic acid is 102.0 whereas the M_r of pentanal is 86.0
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3b

b)

i) The structure of compound **W** is and compound **Z** is formed by:

- Correct structure of compound **W**



ii) Compound **Z** is formed by:

- Compound **Z** is water / H_2O

AND

Produced by a condensation reaction between pentanoic acid and pentan-1-ol / compounds **X** and **Y**; [1 mark]

[Total: 2 marks]

- Compound **X** is pentan-1-ol
 - Straight chain primary alcohol containing five carbon atoms
- You should be aware that the reaction between an alcohol and carboxylic acid produces an ester and water
- In this reaction, concentrated sulfuric acid, H_2SO_4 , is used as a catalyst
- The formation of the ester eliminates a water molecule through a condensation reaction
- **Careful:** A condensation reaction does not necessarily mean that water is formed; it actually means that a small molecule is lost
- The name of compound **W** is pentyl pentanoate

3c

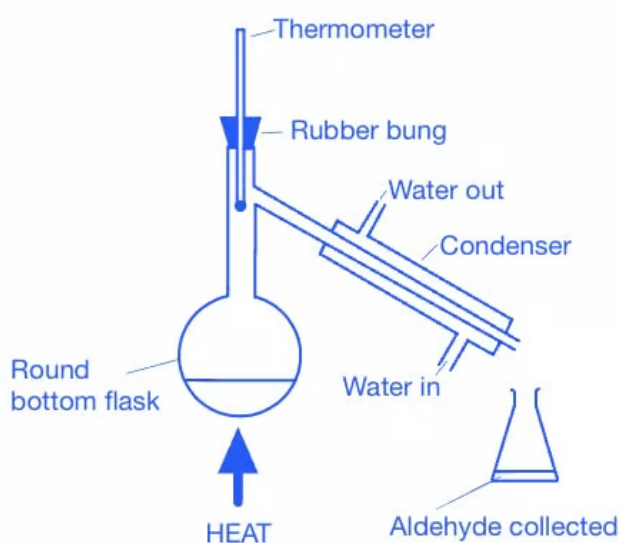
c) A student could prevent the full oxidation of compound **X** by:

- Using an excess of Compound **X** / primary alcohol / pentan-1-ol; [1 mark]

- (To) ensure there are enough Compound **X** / primary alcohol / pentan-1-ol molecules to prevent the second stage of oxidation; [1 mark]
- Distil off aldehyde as soon as it is produced; [1 mark]
- Perform the reaction above the boiling point of the aldehyde / pentanal; [1 mark]

[Total: 4 marks]

- The full oxidation of a primary alcohol such as Compound **X** will produce
 - An aldehyde followed by a carboxylic acid
- If you want to prevent this full oxidation, you **must** alter the reaction conditions and use different apparatus
- A Liebig condenser is used for this purpose
 - This piece of apparatus is ideal for the separation of an organic product from its reacting mixture
 - Allows the distillate to be collected in an approximate boiling point range



3d

d)

i) The structure of the isomer:

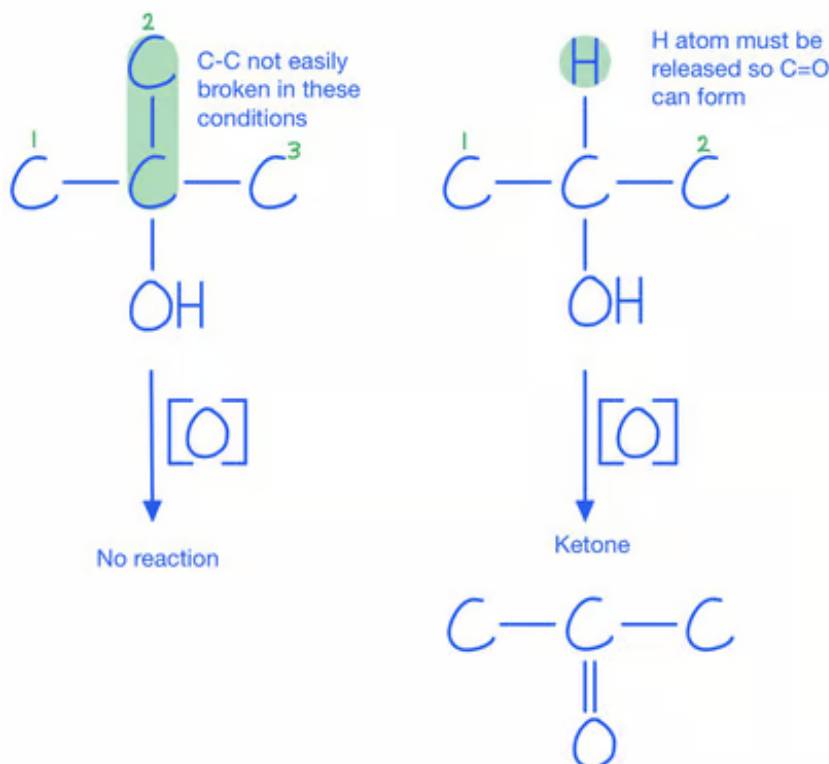
- $\text{CH}_3\text{C}(\text{CH}_3)\text{OHCH}_2\text{CH}_3$; [1 mark]

ii) The reason it can not react with acidified potassium manganate(VII) is:

- Acidified potassium manganate(VII) is an oxidising agent; [1 mark]
- Tertiary alcohols cannot be oxidised; [1 mark]
- To produce an aldehyde / ketone, a C-H must break to allow the C=O bond to form; [1 mark]
- Tertiary alcohols only contain C-C bonds attached to OH group; [1 mark]

[Total: 5 marks]

- This question asks you to think about the difference in not only the primary, secondary and tertiary alcohols but also aldehydes and ketones
- If you are unsure it is best to draw out the molecules



- A primary alcohol will oxidise to form an aldehyde and then a carboxylic acid so the same concept will apply

